PCAP Network Analysis

Problem Statement & Objectives

The problem

Given a PCAP file, show which protocols are present, who talks to whom, and how each TCP connection behaves.

Wireshark is manual, and a label like "retransmission" does not always mean congestion. We show clear numbers and do not present guesses as facts.

Our objectives

1. Upload a PCAP file and click Analyze.
2. Show an overview first, details only on click.
3. Explain TCP flows: RTT, retransmissions, flow control.
4. Show "Not available" instead of fake numbers.

Input

One PCAP or PCAPNG file, uploaded in the browser.

Output

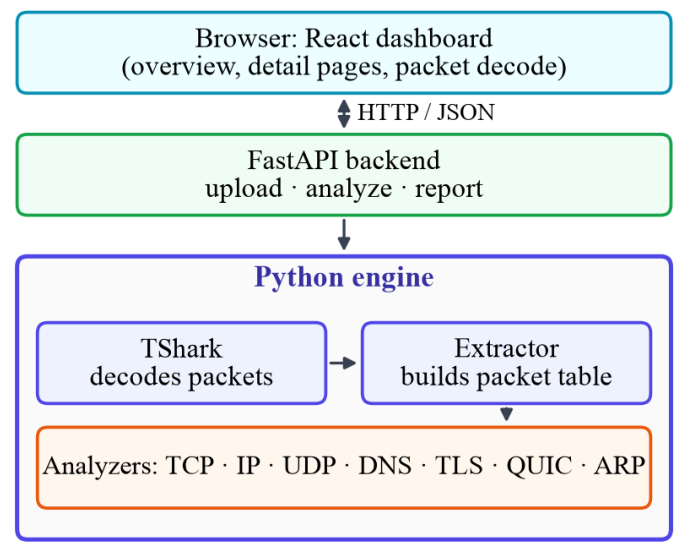
An overview page, detail pages and a JSON report.

Not included

Live capture, decryption, geolocation.

Architecture & How It Works

TShark reads the packets. Our Python code adds up the results.

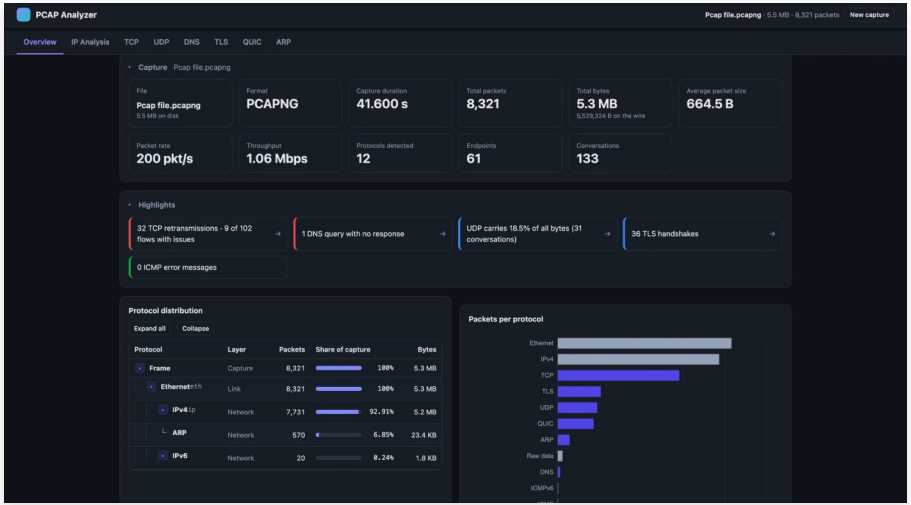


1. **Upload**
   The browser sends the PCAP file to our FastAPI server.
2. **Decode**
   TShark (Wireshark's command-line tool) reads every packet.
3. **Analyze**
   Our Python engine groups packets into flows and does the maths.
4. **Show**
   A React dashboard shows an overview, then detail pages.

The same engine also runs inside two Jupyter notebooks.

Dashboard Flow

The core of the project: from upload to packet details



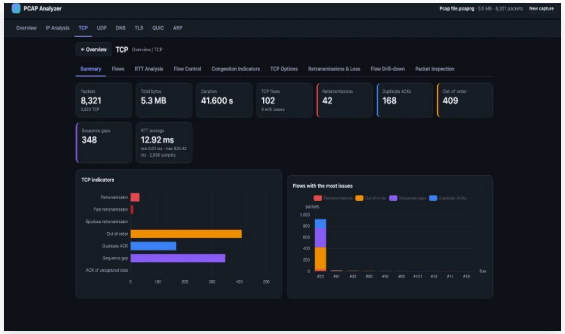
Overview page: capture facts, highlights, protocol tree and chart

Top navbar on every page: Overview · IP Analysis · TCP · UDP · DNS · TLS · QUIC · ARP

1. **Upload**
   Pick a PCAP file and click Analyze.
2. **Overview**
   11 capture facts, highlights, and a chart of all protocols.
3. **Pick a protocol**
   Click a card or the navbar to open its page.
4. **Detail tabs**
   Metrics, charts and tables, split into tabs.
5. **Packet Inspection**
   Click any packet to see its full decode.

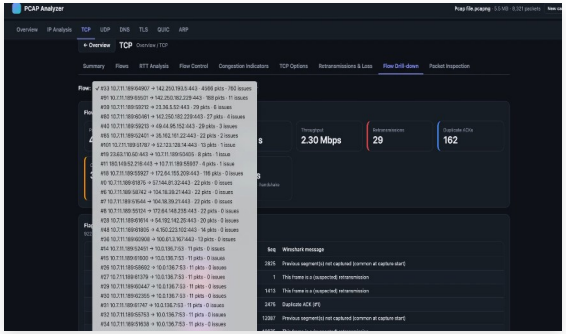
Detail Pages

Detail pages exist for TCP, UDP, DNS, TLS, QUIC, ARP and IP



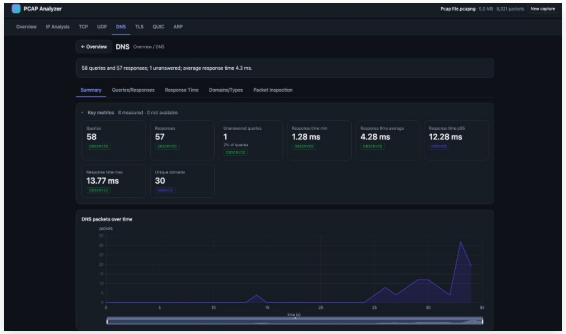
TCP page: 9 tabs

Summary, Flows, RTT, Flow Control, Congestion, Options, Retransmissions, Flow Drill-down, Packet Inspection.



Flow Drill-down

Pick one flow and see its numbers and the Wireshark message for each problem packet.



DNS page: 5 tabs

Queries, response time and domains. Every number is tagged Observed or Derived.

Fixed: ICMP error packets were counted as TCP/UDP; missing values showed 0, now "Not available". **Checked:** 30 of 30 values match when re-counted with separate TShark commands, on 4 sample captures.

Challenges Faced

And how we solved them

1	**TShark field names change**	Names differ between versions. Our code checks which fields exist first.
2	**Big files use a lot of memory**	We read TShark output in chunks, so memory stays near 4 KB per packet.
3	**Some charts were not reliable**	We removed charts we could not trust instead of showing doubtful numbers.